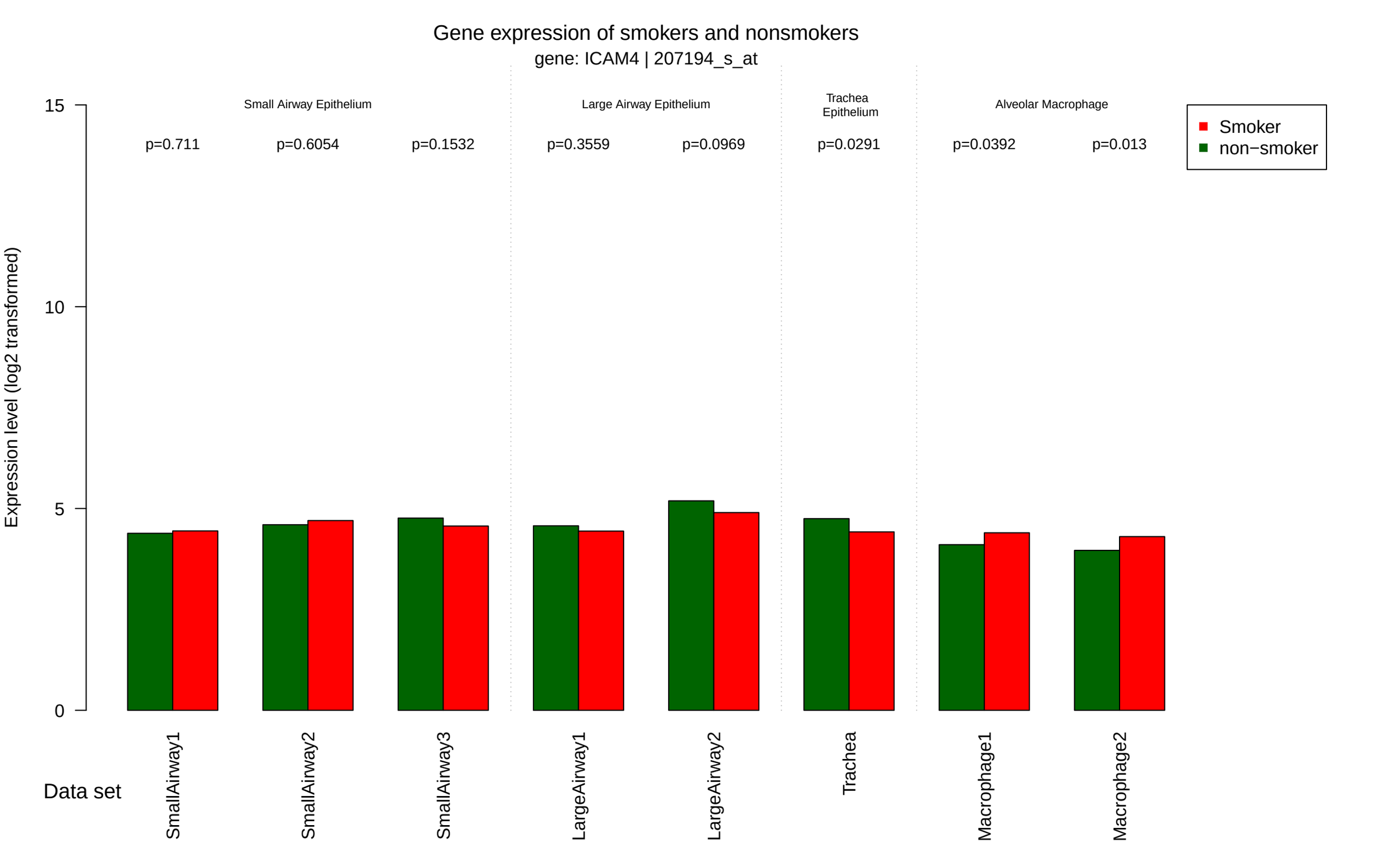
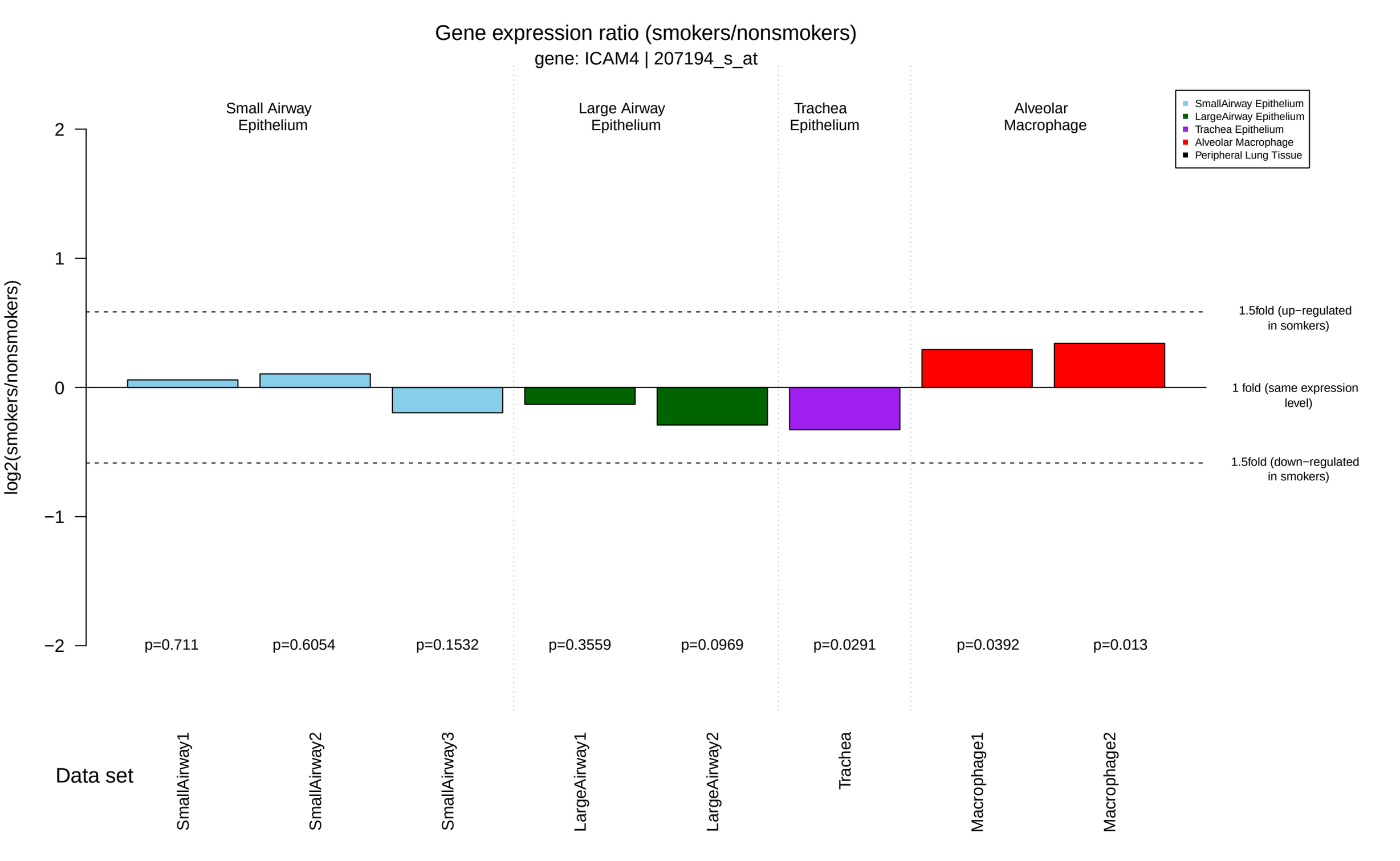
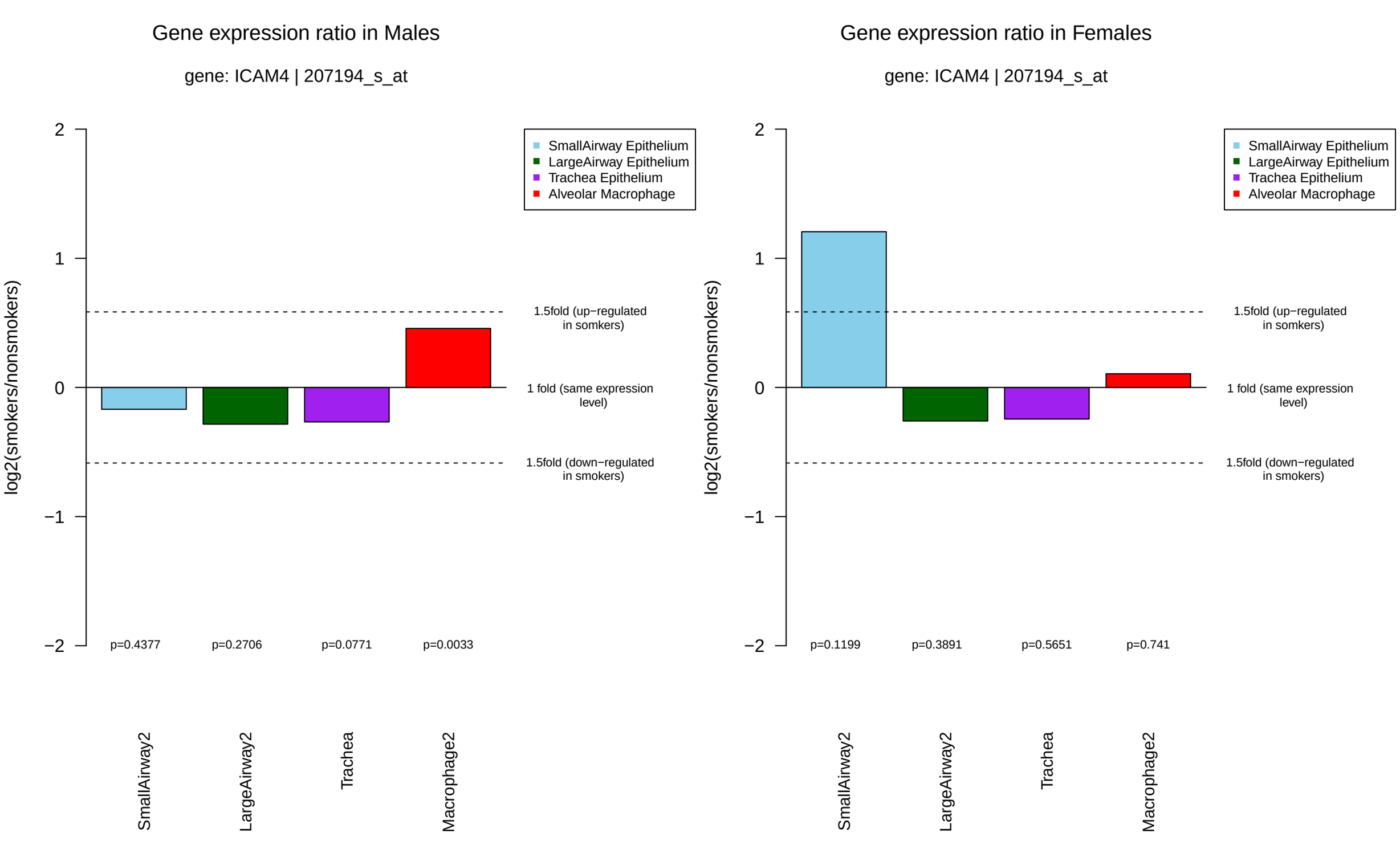
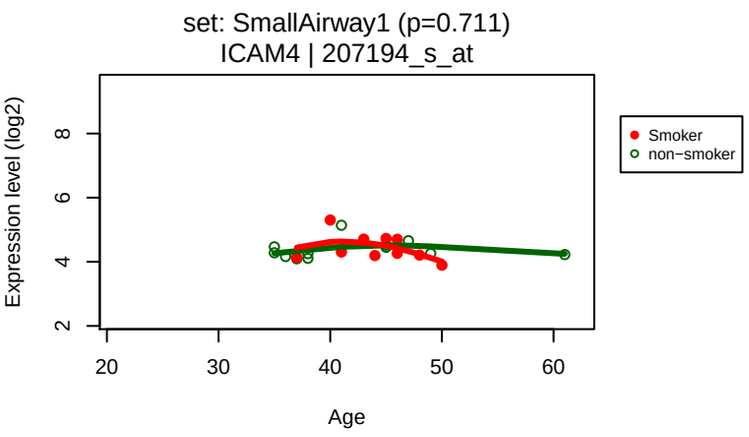
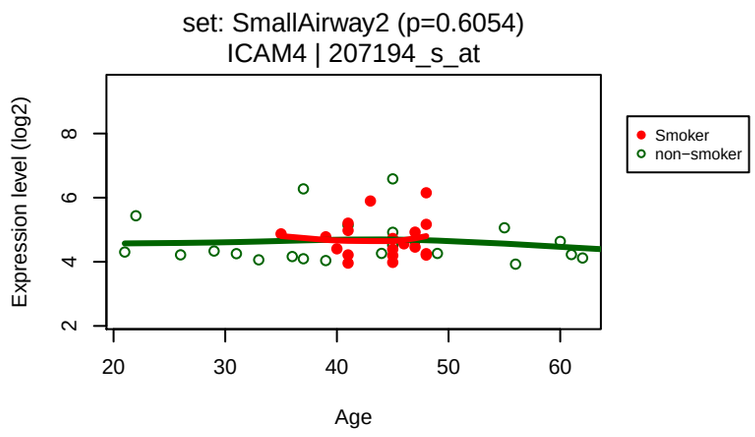
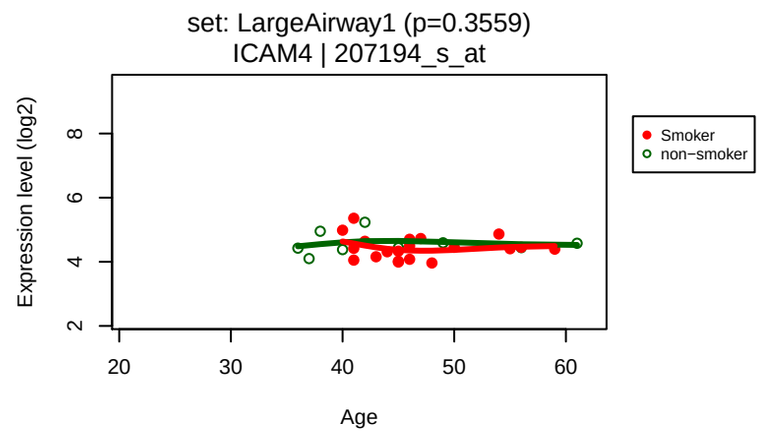

Gene expression of smokers and nonsmokers
gene: ICAM4 | 207194_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.711
p=0.6054
p=0.1532
p=0.3559
p=0.0969
p=0.0291
p=0.0392
p=0.013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

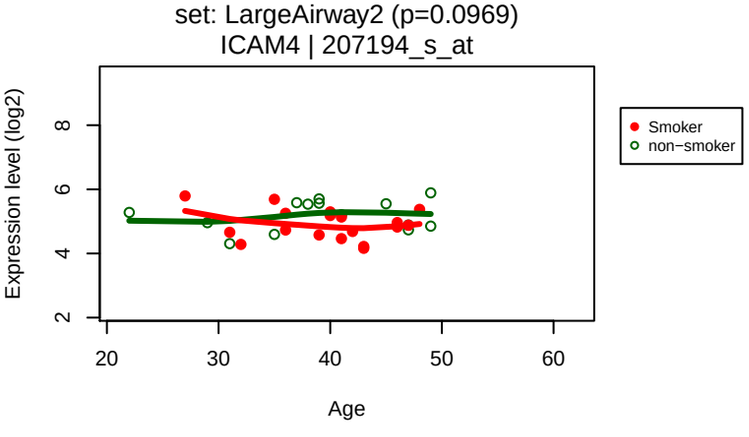

Gene expression ratio (smokers/nonsmokers)
gene: ICAM4 | 207194_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.711
p=0.6054
p=0.1532
p=0.3559
p=0.0969
p=0.0291
p=0.0392
p=0.013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

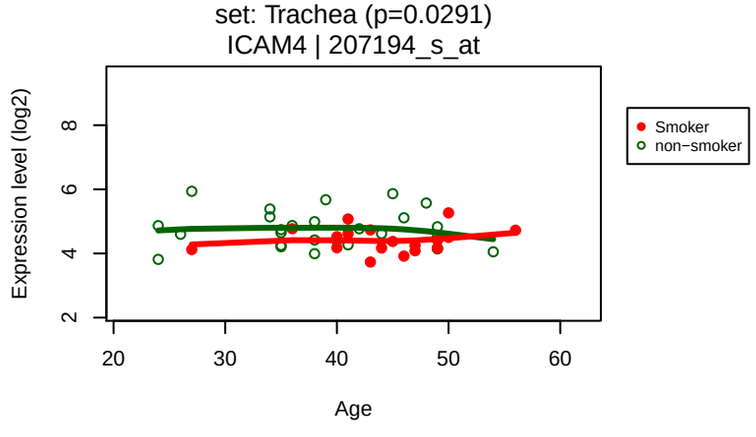

Gene expression ratio in Males
gene: ICAM4 | 207194_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4377
p=0.2706
p=0.0771
p=0.0033
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ICAM4 | 207194_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1199
p=0.3891
p=0.5651
p=0.741
SmallAirway2
LargeAirway2
Trachea
Macrophage2

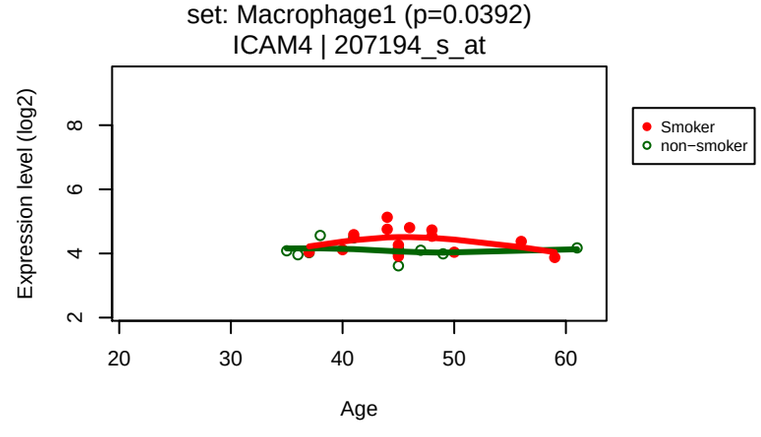

set: SmallAirway1 (p=0.711)
ICAM4 | 207194_s_at
Expression level (log2)
Age
Smoker
non−smoker

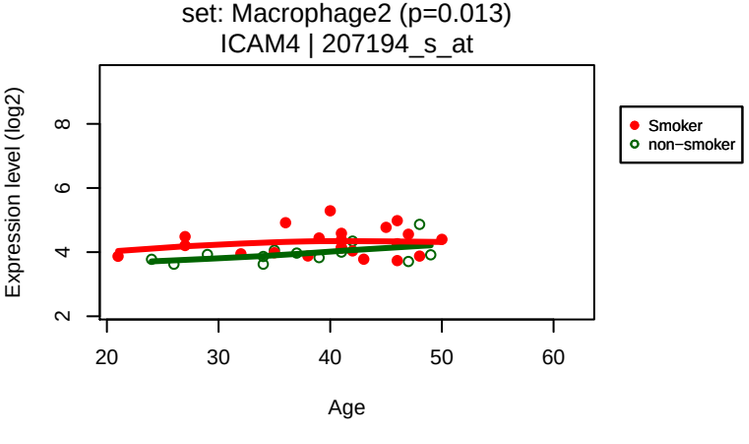

set: SmallAirway2 (p=0.6054)
ICAM4 | 207194_s_at
Expression level (log2)
Age
Smoker
non−smoker

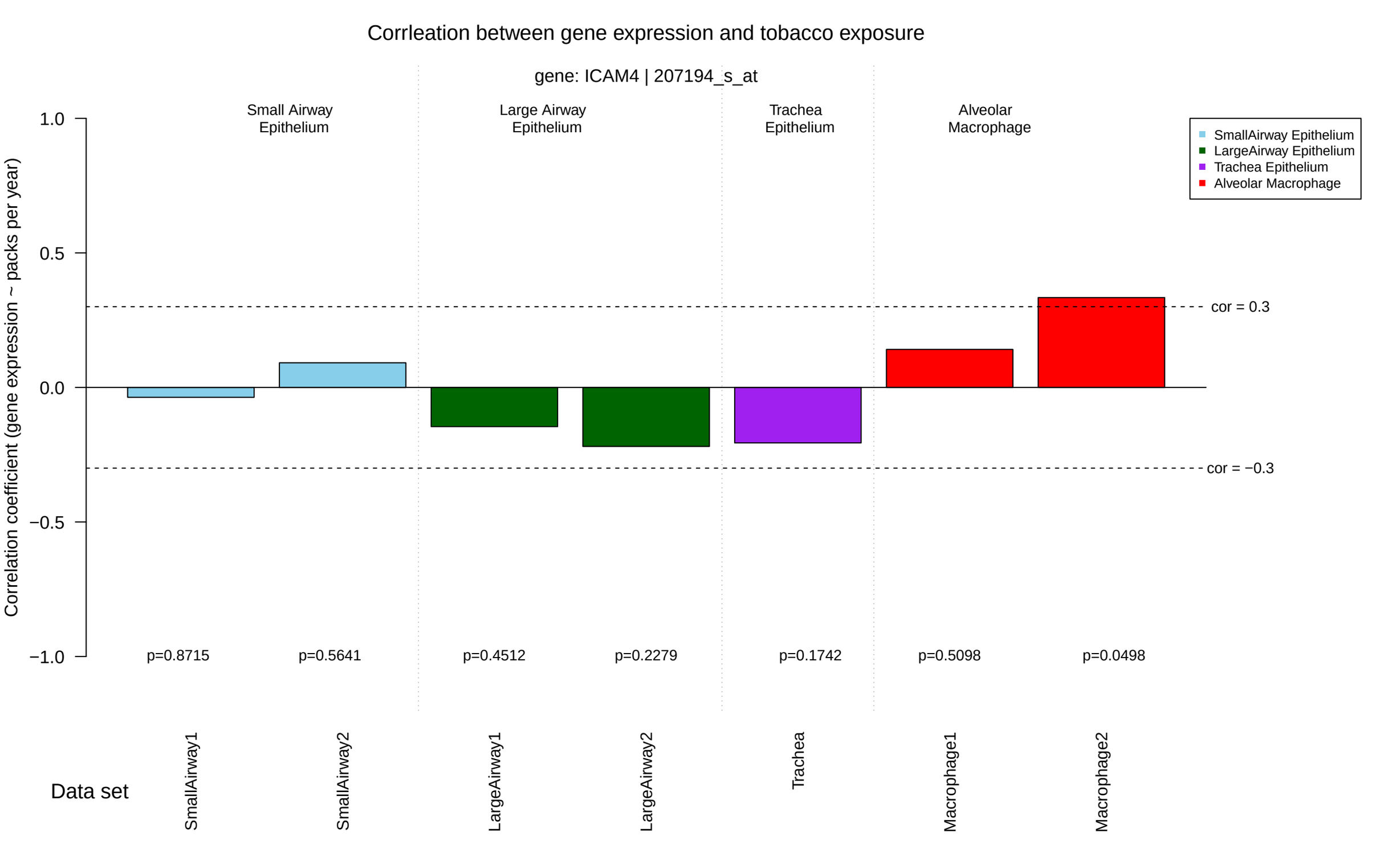

set: LargeAirway1 (p=0.3559)
ICAM4 | 207194_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0969)
ICAM4 | 207194_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0291)
ICAM4 | 207194_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0392)
ICAM4 | 207194_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.013)
ICAM4 | 207194_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ICAM4 | 207194_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8715
p=0.5641
p=0.4512
p=0.2279
p=0.1742
p=0.5098
p=0.0498
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2